CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

Chapter 3: Pair of Linear Equations in Two Variables

Step-by-step solutions to the textbook exercises: 22 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

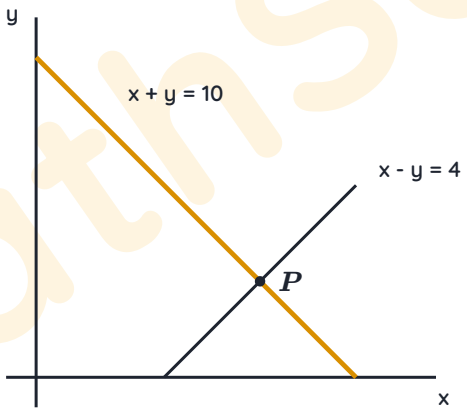
1 Exercise 3.1

Q1 (i) Ten students of Class 10 took part in a quiz. The number of girls is 4 more than the number of boys. Form a pair of linear equations and find the numbers of boys and girls graphically.

SOLUTION

Let the number of girls be x and the number of boys be y. Then $x + y = 10$ and $x - y = 4$.

Points on $x + y = 10$: $(0, 10)$, $(10, 0)$. Points on $x - y = 4$: $(4, 0)$, $(10, 6)$.



The two lines meet at P(7, 3)

The lines meet at $(7, 3)$, so $x = 7$ and $y = 3$.

Answer: 7 girls and 3 boys

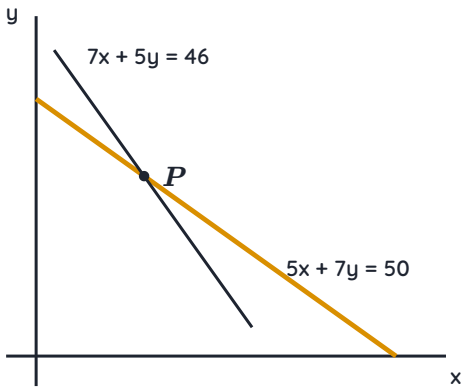
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Q1 (ii) 5 pencils and 7 pens cost ₹50, and 7 pencils and 5 pens cost ₹46. Form a pair of linear equations and find the cost of one pencil and of one pen graphically.

SOLUTION

Let a pencil cost ₹x and a pen cost ₹y. Then $5x + 7y = 50$ and $7x + 5y = 46$.

Points on $5x + 7y = 50$: $(10, 0)$, $(3, 5)$. Points on $7x + 5y = 46$: $(3, 5)$, $(8, -2)$.



The two lines meet at P(3, 5)

The lines meet at $(3, 5)$, so $x = 3$ and $y = 5$. Check: $5(3) + 7(5) = 50$ and $7(3) + 5(5) = 46$.

Answer: a pencil costs ₹3 and a pen costs ₹5

Q2 By comparing $\frac{a_1}{a_2}$, $\frac{b_1}{b_2}$ and $\frac{c_1}{c_2}$, say whether the lines intersect at a point, are parallel or coincide: (i) $5x - 4y + 8 = 0$, $7x + 6y - 9 = 0$ (ii) $9x + 3y + 12 = 0$, $18x + 6y + 24 = 0$ (iii) $6x - 3y + 10 = 0$, $2x - y + 9 = 0$.

SOLUTION

(i) $\frac{a_1}{a_2} = \frac{5}{7}$ and $\frac{b_1}{b_2} = \frac{-4}{6} = -\frac{2}{3}$. These are not equal, so the lines intersect at a point.

(ii) $\frac{9}{18} = \frac{3}{6} = \frac{12}{24} = \frac{1}{2}$. All three ratios are equal, so the lines coincide.

(iii) $\frac{6}{2} = 3$, $\frac{-3}{-1} = 3$ and $\frac{10}{9}$. The first two ratios are equal but the third is different, so the lines are parallel.

Answer: (i) intersecting (ii) coincident (iii) parallel

Q3 By comparing the ratios, say whether each pair is consistent or inconsistent: (i) $3x + 2y = 5, 2x - 3y = 7$ (ii) $2x - 3y = 8, 4x - 6y = 9$ (iii) $\frac{3}{2}x + \frac{5}{3}y = 7, 9x - 10y = 14$ (iv) $5x - 3y = 11, -10x + 6y = -22$ (v) $\frac{4}{3}x + 2y = 8, 2x + 3y = 12$.

SOLUTION

A pair is consistent if it has at least one solution (intersecting or coincident lines) and inconsistent if it has none (parallel lines).

(i) $\frac{3}{2} \neq \frac{2}{-3}$, so the lines intersect. Consistent.

(ii) $\frac{2}{4} = \frac{-3}{-6} = \frac{1}{2}$ but $\frac{8}{9} \neq \frac{1}{2}$, so the lines are parallel. Inconsistent.

(iii) $\frac{3/2}{9} = \frac{1}{6}$ and $\frac{5/3}{-10} = -\frac{1}{6}$. These are not equal, so the lines intersect. Consistent.

(iv) $\frac{5}{-10} = \frac{-3}{6} = \frac{11}{-22} = -\frac{1}{2}$, so the lines coincide. Consistent, with infinitely many solutions.

(v) $\frac{4/3}{2} = \frac{2}{3}, \frac{2}{3}$ and $\frac{8}{12} = \frac{2}{3}$, so the lines coincide. Consistent, with infinitely many solutions.

Answer: (i) consistent (ii) inconsistent (iii) consistent (iv) consistent (v) consistent

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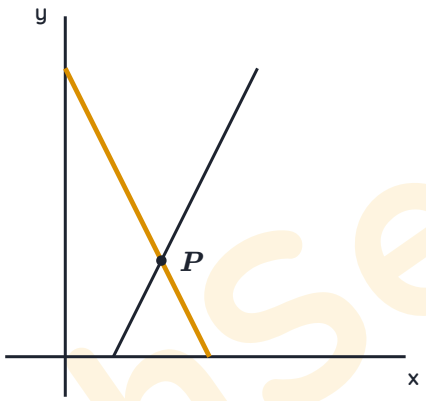
Q4 Which of these pairs are consistent? Solve the consistent ones graphically: (i) $x + y = 5, 2x + 2y = 10$ (ii) $x - y = 8, 3x - 3y = 16$ (iii) $2x + y - 6 = 0, 4x - 2y - 4 = 0$ (iv) $2x - 2y - 2 = 0, 4x - 4y - 5 = 0$.

SOLUTION

(i) $\frac{1}{2} = \frac{1}{2} = \frac{5}{10}$, so the lines coincide and the pair is consistent. Both equations give the same line through $(0, 5)$ and $(5, 0)$. Every point on it, such as $(1, 4)$ or $(2, 3)$, is a solution, so there are infinitely many solutions.

(ii) $\frac{1}{3} = \frac{-1}{-3}$ but $\frac{8}{16} = \frac{1}{2}$ is different, so the lines are parallel. Inconsistent.

(iii) $\frac{2}{4} \neq \frac{1}{-2}$, so the lines intersect and the pair is consistent. Points on $2x + y = 6$: $(0, 6)$, $(3, 0)$. Points on $4x - 2y = 4$: $(1, 0)$, $(4, 6)$.



The lines 2x + y = 6 and 4x - 2y = 4 meet at P(2, 2)

The lines meet at $(2, 2)$, so $x = 2, y = 2$.

(iv) $\frac{2}{4} = \frac{-2}{-4} = \frac{1}{2}$ but $\frac{-2}{-5} = \frac{2}{5}$ is different, so the lines are parallel. Inconsistent.

Answer: (i) consistent, infinitely many solutions (ii) inconsistent (iii) consistent, $x = 2, y = 2$ (iv) inconsistent

Q5 Half the perimeter of a rectangular garden is 36 m, and its length is 4 m more than its width. Find its dimensions.

SOLUTION

Let the width be x m and the length be y m. Then $y = x + 4$ and $x + y = 36$.

Substituting, $x + x + 4 = 36$, so $x = 16$ and $y = 20$.

Answer: length 20 m and width 16 m

Q6 For the equation $2x + 3y - 8 = 0$, write another linear equation so that the pair represents (i) intersecting lines (ii) parallel lines (iii) coincident lines.

SOLUTION

(i) Choose coefficients with $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, for example $3x + 2y - 7 = 0$ $\left(\frac{2}{3} \neq \frac{3}{2}\right)$.

(ii) Keep the ratio of the coefficients the same but change the constant, for example $4x + 6y - 12 = 0$ $\left(\frac{2}{4} = \frac{3}{6} \neq \frac{8}{12}\right)$.

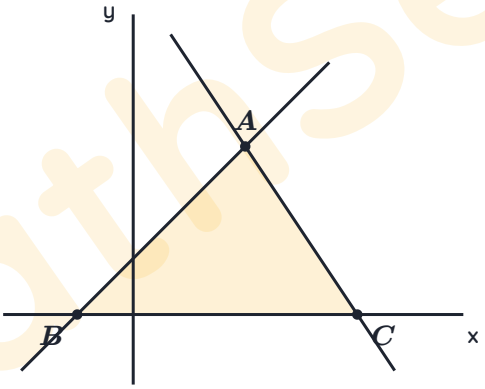
(iii) Multiply the whole equation by a number, for example $4x + 6y - 16 = 0$.

Answer: (i) $3x + 2y - 7 = 0$ (ii) $4x + 6y - 12 = 0$ (iii) $4x + 6y - 16 = 0$. Other answers are possible.

Q7 Draw the graphs of $x - y + 1 = 0$ and $3x + 2y - 12 = 0$. Find the vertices of the triangle formed by these lines and the x-axis, and shade the triangle.

SOLUTION

Points on $x - y + 1 = 0$: $(-1, 0)$, $(0, 1)$, $(2, 3)$. Points on $3x + 2y - 12 = 0$: $(4, 0)$, $(2, 3)$, $(0, 6)$.



The shaded triangle has vertices A(2, 3), B(-1, 0) and C(4, 0)

The two lines meet at $(2, 3)$. The first line cuts the x-axis at $(-1, 0)$ and the second cuts it at $(4, 0)$.

Answer: the vertices are $(2, 3)$, $(-1, 0)$ and $(4, 0)$

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2 Exercise 3.2

Q1 Solve by the substitution method: (i) $x + y = 14$, $x - y = 4$ (ii) $s - t = 3$, $\frac{s}{3} + \frac{t}{2} = 6$ (iii) $3x - y = 3$, $9x - 3y = 9$ (iv) $0.2x + 0.3y = 1.3$, $0.4x + 0.5y = 2.3$ (v) $\sqrt{2}\,x + \sqrt{3}\,y = 0$, $\sqrt{3}\,x - \sqrt{8}\,y = 0$ (vi) $\frac{3x}{2} - \frac{5y}{3} = -2$, $\frac{x}{3} + \frac{y}{2} = \frac{13}{6}$.

SOLUTION

(i) From the second equation, $x = y + 4$. Then $y + 4 + y = 14$, so $y = 5$ and $x = 9$.

(ii) $s = t + 3$. Then $\frac{t+3}{3} + \frac{t}{2} = 6$ gives $2t + 6 + 3t = 36$, so $t = 6$ and $s = 9$.

(iii) $y = 3x - 3$. Then $9x - 3(3x - 3) = 9$ gives $9 = 9$, which is always true. The two equations are the same line, so there are infinitely many solutions.

(iv) Multiply both by 10: $2x + 3y = 13$ and $4x + 5y = 23$. From the first, $x = \frac{13 - 3y}{2}$. Then $2(13 - 3y) + 5y = 23$ gives $y = 3$ and $x = 2$.

(v) From the first, $x = -\frac{\sqrt{3}}{\sqrt{2}}\,y$. Then $-\frac{3}{\sqrt{2}}\,y - 2\sqrt{2}\,y = 0$, so $y = 0$ and $x = 0$.

(vi) Clearing fractions: $9x - 10y = -12$ and $2x + 3y = 13$. From the second, $x = \frac{13 - 3y}{2}$. Then $9(13 - 3y) - 20y = -24$ gives $47y = 141$, so $y = 3$ and $x = 2$.

Answer: (i) $x = 9$, $y = 5$ (ii) $s = 9$, $t = 6$ (iii) infinitely many solutions (iv) $x = 2$, $y = 3$ (v) $x = 0$, $y = 0$ (vi) $x = 2$, $y = 3$

Q2 Solve $2x + 3y = 11$ and $2x - 4y = -24$, and hence find m for which $y = mx + 3$.

SOLUTION

From the first equation, $2x = 11 - 3y$. Substituting in the second: $11 - 3y - 4y = -24$, so $7y = 35$ and $y = 5$.

Then $2x = 11 - 15 = -4$, so $x = -2$.

In $y = mx + 3$: $5 = -2m + 3$, so $m = -1$.

Answer: $x = -2$, $y = 5$ and $m = -1$

Q3 (i) Two numbers differ by 26, and one of them is three times the other. What are the numbers?

SOLUTION

Let the numbers be x and y with $x > y$. Then $x - y = 26$ and $x = 3y$.

Substituting, $3y - y = 26$, so $y = 13$ and $x = 39$.

Answer: 39 and 13

Q3 (ii) Two angles are supplementary, and the bigger one is 18° more than the smaller one. Find both angles.

SOLUTION

Let the angles be x° and y° with $x > y$. Then $x + y = 180$ and $x = y + 18$.

Substituting, $2y + 18 = 180$, so $y = 81$ and $x = 99$.

Answer: 99° **and** 81°

Q3 (iii) 7 bats and 6 balls cost ₹3800, and 3 bats and 5 balls cost ₹1750. Find the cost of each bat and each ball.

SOLUTION

Let a bat cost ₹x and a ball cost ₹y. Then $7x + 6y = 3800$ and $3x + 5y = 1750$.

From the second, $x = \dfrac{1750 - 5y}{3}$. Then $7(1750 - 5y) + 18y = 11400$ gives $17y = 850$, so $y = 50$ and $x = \dfrac{1750 - 250}{3} = 500$.

Answer: a bat costs ₹500 and a ball costs ₹50

Q3 (iv) A taxi charges a fixed amount plus a charge per km. A 10 km journey costs ₹105 and a 15 km journey costs ₹155. Find the fixed charge and the charge per km. What does a 25 km journey cost?

SOLUTION

Let the fixed charge be ₹x and the charge per km be ₹y. Then $x + 10y = 105$ and $x + 15y = 155$.

From the first, $x = 105 - 10y$. Then $105 - 10y + 15y = 155$, so $y = 10$ and $x = 5$.

For 25 km: $5 + 25 \times 10 = ₹255$.

Answer: fixed charge ₹5, ₹10 per km, and ₹255 for 25 km

Q3 (v) A fraction becomes $\frac{9}{11}$ if 2 is added to both the numerator and the denominator, and $\frac{5}{6}$ if 3 is added to both. Find the fraction.

SOLUTION

Let the fraction be $\dfrac{x}{y}$. Then $\dfrac{x+2}{y+2} = \dfrac{9}{11}$ gives $11x - 9y = -4$, and $\dfrac{x+3}{y+3} = \dfrac{5}{6}$ gives $6x - 5y = -3$.

From the second, $x = \dfrac{5y - 3}{6}$. Then $11(5y - 3) - 54y = -24$ gives $y = 9$ and $x = 7$.

Answer: $\dfrac{7}{9}$

Q3 (vi) Five years from now a father will be three times as old as his son. Five years ago he was seven times as old as his son. Find their present ages.

SOLUTION

Let the father be x years and the son y years old. Then $x+5=3(y+5)$ gives $x-3y=10$, and $x-5=7(y-5)$ gives $x-7y=-30$.

From the first, $x=3y+10$. Then $3y+10-7y=-30$, so $y=10$ and $x=40$.

Answer: the father is 40 years old and the son is 10 years old

3 Exercise 3.3

Q1 Solve by the elimination method and by the substitution method: (i) $x+y=5, 2x-3y=4$ (ii) $3x+4y=10, 2x-2y=2$ (iii) $3x-5y-4=0, 9x=2y+7$ (iv) $\frac{x}{2}+\frac{2y}{3}=-1, x-\frac{y}{3}=3$.

SOLUTION

(i) Elimination: multiply the first by 3 to get $3x+3y=15$ and add the second: $5x=19$, so $x=\frac{19}{5}$ and $y=5-\frac{19}{5}=\frac{6}{5}$. Substitution: $x=5-y$ gives $10-2y-3y=4$, so $y=\frac{6}{5}$, the same result.

(ii) Elimination: multiply the second by 2 to get $4x-4y=4$ and add the first: $7x=14$, so $x=2$ and $y=1$. Substitution: $x=y+1$ gives $3y+3+4y=10$, so $y=1$.

(iii) The equations are $3x-5y=4$ and $9x-2y=7$. Elimination: multiply the first by 3 to get $9x-15y=12$ and subtract the second: $-13y=5$, so $y=-\frac{5}{13}$ and $x=\frac{4+5y}{3}=\frac{9}{13}$. Substitution: $y=\frac{9x-7}{2}$ gives $6x-45x+35=8$, so $x=\frac{9}{13}$.

(iv) Clearing fractions: $3x+4y=-6$ and $3x-y=9$. Elimination: subtracting, $5y=-15$, so $y=-3$ and $x=2$. Substitution: $y=3x-9$ gives $3x+12x-36=-6$, so $x=2$.

Answer: (i) $x=\frac{19}{5}, y=\frac{6}{5}$ **(ii)** $x=2, y=1$ **(iii)** $x=\frac{9}{13}, y=-\frac{5}{13}$ **(iv)** $x=2, y=-3$

Q2 (i) A fraction becomes 1 if 1 is added to the numerator and 1 is subtracted from the denominator. It becomes $\frac{1}{2}$ if 1 is added to the denominator only. Find the fraction.

SOLUTION

Let the fraction be $\frac{x}{y}$. Then $x+1=y-1$ gives $x-y=-2$, and $2x=y+1$ gives $2x-y=1$.

Subtracting the first from the second, $x=3$. Then $y=5$.

Answer: $\frac{3}{5}$

Q2 (ii) Five years ago one person was three times as old as another. Ten years from now the first will be twice as old as the second. Find their present ages.

SOLUTION

Let their ages be x and y years. Then $x - 5 = 3(y - 5)$ gives $x - 3y = -10$, and $x + 10 = 2(y + 10)$ gives $x - 2y = 10$.

Subtracting the first from the second, $y = 20$. Then $x = 50$.

Answer: 50 years and 20 years

Q2 (iii) The sum of the digits of a two-digit number is 9. Nine times the number is twice the number obtained by reversing its digits. Find the number.

SOLUTION

Let the tens digit be x and the units digit be y. Then $x + y = 9$, and $9(10x + y) = 2(10y + x)$ gives $88x - 11y = 0$, that is $8x - y = 0$.

Adding the two equations, $9x = 9$, so $x = 1$ and $y = 8$.

Answer: 18

Q2 (iv) A person withdraws ₹2000 from a bank in ₹50 and ₹100 notes only, and gets 25 notes in all. How many notes of each kind are there?

SOLUTION

Let there be x notes of ₹50 and y notes of ₹100. Then $x + y = 25$, and $50x + 100y = 2000$ gives $x + 2y = 40$.

Subtracting the first from the second, $y = 15$. Then $x = 10$.

Answer: 10 notes of ₹50 and 15 notes of ₹100

Q2 (v) A library has a fixed charge for the first three days and an extra charge for each day after that. One reader paid ₹27 for seven days and another paid ₹21 for five days. Find the fixed charge and the charge for each extra day.

SOLUTION

Let the fixed charge be ₹x and the charge per extra day be ₹y. Seven days means 4 extra days and five days means 2 extra days: $x + 4y = 27$ and $x + 2y = 21$.

Subtracting, $2y = 6$, so $y = 3$ and $x = 15$.

Answer: fixed charge ₹15 and ₹3 for each extra day

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